

Sutton and Barton Exercises Chapter 11

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1 Chapter 11

Exercise 11.1 Convert the equation of n -step on-policy TD (7.9) to semi-gradient form. Give accompanying definitions of the return for both the episodic and continuing cases.

We are pretty much just rewriting things in terms of function approximation...

$$\mathbf{w}_{t+n} \leftarrow \mathbf{w}_{t+n-1} + \alpha \rho_{t:t+n-1} [G_{t:t+n} - \hat{v}(S_t, \mathbf{w}_{t+n-1})] \nabla \hat{v}(S_t, \mathbf{w}_{t+n-1})$$

where the return is defined as:

Episodic:

$$G_{t:t+n} \doteq R_{t+1} + \gamma R_{t+2} + \cdots + \gamma^{n-1} R_{t+n} + \gamma^n \hat{v}(S_{t+n}, \mathbf{w}_{t+n-1})$$

Continuing:

$$G_{t:t+n} \doteq R_{t+1} - \bar{R}_t + R_{t+2} - \bar{R}_{t+1} + \cdots + R_{t+n} - \bar{R}_{t+n-1} + \hat{v}(S_{t+n}, \mathbf{w}_{t+n-1})$$

Exercise 11.2 The semi-gradient form of n -step $Q(\sigma)$ is:

$$\mathbf{w}_{t+n} \leftarrow \mathbf{w}_{t+n-1} + \alpha [G_{t:t+n} - \hat{q}(S_t, A_t, \mathbf{w}_{t+n-1})] \nabla \hat{q}(S_t, A_t, \mathbf{w}_{t+n-1})$$

where the $Q(\sigma)$ return is:

Episodic:

$$G_{t:t+n} \doteq \hat{q}(S_t, A_t, \mathbf{w}) + \sum_{k=t}^{t+n-1} \delta_k \prod_{i=t+1}^k (\gamma \sigma_i \rho_i + \gamma(1 - \sigma_i) \pi(A_i | S_i))$$

Continuing:

$$G_{t:t+n} \doteq \hat{q}(S_t, A_t, \mathbf{w}) + \sum_{k=t}^{t+n-1} \delta_k \prod_{i=t+1}^k \left(\gamma \sigma_i \rho_i + \gamma(1 - \sigma_i) \pi(A_i | S_i) \right)$$

where in the continuing case the TD error uses the differential form:

$$\delta_k = R_{k+1} - \bar{R} + \hat{q}(S_{k+1}, A_{k+1}, \mathbf{w}) - \hat{q}(S_k, A_k, \mathbf{w})$$

Exercise 11.4 Prove (11.24).:

Derivation:

Start with the definition of $\overline{RE}$:

$$\overline{RE}(\mathbf{w}) = E \left[(G_t - \hat{v}(S_t, \mathbf{w}))^2 \right]$$

Add and subtract $v_\pi(S_t)$ inside the bracket:

$$= E \left[(G_t - v_\pi(S_t) + v_\pi(S_t) - \hat{v}(S_t, \mathbf{w}))^2 \right]$$

Let $a = G_t - v_\pi(S_t)$ and $b = v_\pi(S_t) - \hat{v}(S_t, \mathbf{w})$, expand $(a + b)^2$:

$$= E \left[(G_t - v_\pi(S_t))^2 \right] + 2E \left[(G_t - v_\pi(S_t))(v_\pi(S_t) - \hat{v}(S_t, \mathbf{w})) \right] + E \left[(v_\pi(S_t) - \hat{v}(S_t, \mathbf{w}))^2 \right]$$

The cross term vanishes because $E[G_t - v_\pi(S_t) \mid S_t] = 0$ by definition of $v_\pi(S_t) = E[G_t \mid S_t]$:

$$E \left[(G_t - v_\pi(S_t))(v_\pi(S_t) - \hat{v}(S_t, \mathbf{w})) \right] = E \left[E[G_t - v_\pi(S_t) \mid S_t] \cdot (v_\pi(S_t) - \hat{v}(S_t, \mathbf{w})) \right] = 0$$

What remains:

$$\overline{RE}(\mathbf{w}) = E \left[(v_\pi(S_t) - \hat{v}(S_t, \mathbf{w}))^2 \right] + E \left[(G_t - v_\pi(S_t))^2 \right] = \overline{VE}(\mathbf{w}) + E \left[(G_t - v_\pi(S_t))^2 \right]$$

The second term is the variance of G_t around the true value $v_\pi(S_t)$ and does not depend on $\mathbf{w}$. Therefore $\overline{RE}$ and $\overline{VE}$ share the same optimal parameter vector $\mathbf{w}^*$.